

Cooperative Multilateral Tariff Negotiations

A Quantitative Analysis Using Shapley Value

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Multilateral tariff bargaining is the modal case

- Major trade pacts of the last 30 years are *multilateral* cooperative bargains:

NAFTA / USMCA	3 members
EU customs union	27
RCEP	15
CPTPP	11
WTO Doha (Swiss formula)	~160
- The quantitative literature on tariff bargaining is overwhelmingly *bilateral*: Bagwell–Staiger–Yurukoglu (2021, BSY); Bagwell–Staiger reciprocity tradition.
- Two-party Nash bargaining $\rightarrow$ exogenous bargaining weights μ_i, μ_j estimated from observed RTAs.

Question. What is the right structural surplus-allocation rule when $N > 2$ countries cooperatively cut tariffs?

Bilateral → multilateral: a substantive new modeling choice

Bilateral (BSY 2021):

$$\max_{\tau} (W_i(\tau) - W_i^0)^{\mu_i} (W_j(\tau) - W_j^0)^{\mu_j}, \quad \mu_i + \mu_j = 1$$

Multilateral $N > 2$:

$$\max_{\tau_S} \prod_{n \in S} (W_n(\tau_S) - W_n^0)^{\sigma_n}, \quad \sum_{n \in S} \sigma_n = 1$$

- How to specify σ_n ? GDP-weighted? Population-weighted? Trade-volume-weighted?
- Each ad-hoc choice gives a different surplus allocation, with no internally consistent criterion.
- Need a single rule that ties σ to coalition-level marginal contributions.

This paper

What we do. Combine the Caliendo–Parro (2015) GE primitives with the Shapley value as a structural *imputation rule* for cooperative tariff surplus. Extend BSY (2021) from $N = 2$ to $N = 7$.

Three contributions.

1. **Multilateral cooperative bargaining** that nests BSY's symmetric Nash bargain in the $N = 2$ limit.
2. **Single-step MPEC** solve closes a 6.8% efficiency gap silently lost by BSY's two-step heuristic on NAFTA.
3. **Headline finding:** Shapley shares are *uncorrelated* with GDP shares (Pearson $\in [-0.26, +0.01]$); they track *tariff-liberalization headroom*.

What we do not claim. Identification of real-world bargaining power. The framework is a structural imputation rule under four explicit modeling choices.

Where this paper sits

	GE	Solution	N	Tariff lever
Bagwell–Staiger–Yurukoglu (2021)	EK-CP	Nash bargain	2	sectoral
Caliendo–Parro (2015)	EK-CP	none (calib)	31	none
Ossa (2014, AER)	Krugman	Nash	7	sectoral
Agnew (2023, 2025)	1-sector	Shapley	≤ 3	uniform
Limaο (2007)	analyt.	–	–	PTA only
Chander–Tulkens (γ -core)	–	γ -core	any	–
This paper	EK-CP	Shapley	≤ 7 exact	scalar s_n

Distinct combination:

- Multi-sector EK-CP $\times$ Shapley value (vs. Agnew's 1-sector toy).
- Single-step MPEC vs. BSY two-step (welfare gap quantified).
- Honest scope: not bargaining-power identification.

Model: Caliendo–Parro hat-algebra GE

Trade environment.

- N countries $\times J$ sectors. Eaton–Kortum production with intermediate-input I-O linkages $\gamma_n^{k,j}$.
- Frechet productivity, shape θ^j . Bilateral iceberg d_{ni}^j and tariff τ_{ni}^j , so effective wedge $\hat{\kappa}_{ni}^j = (1 + \tau_{ni}^{j'}) / (1 + \tau_{ni}^j)$.
- Hat-algebra: equilibrium in changes depends only on *observed* base-year shares $\{\pi_{ni}^j, X_n^j\}$ + elasticities θ^j .

Welfare. $\widehat{W}_n = \widehat{I}_n / \prod_j (\widehat{P}_n^j)^{\alpha_n^j}$ (ACR aggregation; Arkolakis-Costinot-Rodriguez-Clare 2012).

Coalitional value. For coalition S , with σ_S Nash weights,

$$v(S \mid \sigma) = \sum_{n \in S} I_n^{\text{base}} (\widehat{W}_n^* - 1), \quad \widehat{W}^* \in \arg \max_{\tau_S} \prod_{n \in S} (\widehat{W}_n - 1)^{\sigma_n}.$$

Lever. $\tau_{ni}^{j,\text{new}} = s_n \tau_{ni}^{j,0}$, $s_n \in [0, 1]$ (stylized uniform proportional cut).

Single-step MPEC formulation

$$\begin{aligned} & \max_{\substack{\{s_n\}_{n \in S}, \\ \widehat{w}, \widehat{P}, \widehat{c}, \widehat{\pi}, \widehat{X}'}} \sum_{i \in S} \sigma_i \log(\widehat{W}_i - 1) \\ & \text{s.t. [5 equilibrium equations of CP 2015 hat-algebra]} \\ & \quad \widehat{\kappa}_{ni}^j(s_n) = (1 + s_n \tau_{ni}^{j,0}) / (1 + \tau_{ni}^{j,0}) \\ & \quad \widehat{W}_i \geq 1 + \epsilon \quad \forall i \in S \quad (\text{IR}) \\ & \quad s_n \in [0, 1] \quad \forall n \in S; \quad s_n = 1 \quad \forall n \notin S. \end{aligned}$$

Two ways to solve:

- BSY-style **two-step**: approximate inner GE; fast but lossy.
- Our **single-step**: full hat-algebra at every outer iter.

The trade-off matters – benchmarked in slide 13:

\$5.49B $\rightarrow$ \$5.89B on NAFTA grand coalition (+6.8%), at the cost of $\sim 2014\times$ wall-clock.

Solution concept: Shapley value with a self-referential weight map

Shapley value (Shapley 1953).

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (N - |S| - 1)!}{N!} [v(S \cup \{i\}) - v(S)]$$

Why Shapley over alternatives?

- **Nests BSY at $N = 2$:** $\sigma_i^* = \sigma_j^* = \frac{1}{2}$ recovers the symmetric Nash bargain (Proposition 1, see backup).
- **Unique** allocation satisfying efficiency, symmetry, dummy, additivity (Hart-Mas-Colell 1989).
- Closed-form computability (vs. nucleolus, γ -core).

Self-referential map. Because v depends on σ , define

$$\sigma^{(k+1)} = (1 - \alpha) \sigma^{(k)} + \alpha \frac{\phi(v(\cdot \mid \sigma^{(k)}))}{\sum_n \phi_n(\cdot)}, \quad \alpha = 0.5.$$

At $N = 3$: 3 random Dirichlet starts converge to spread $< 2 \times 10^{-2}$ in 5 iters (numerical

Data: Caliendo–Parro (2015) panel

Source. Caliendo–Parro (2015, RES) replication archive.

- **30 countries** + Rest-of-World residual.
- **40 sectors** (20 manufacturing, 20 services).
- Base year 1993, calibrated to bilateral trade shares + I-O.
- Trade elasticities θ^j from CP 2015 Table 4 (Caliendo–Parro estimates from triple-differenced bilateral trade gravity).

$N = 3$: USA, Mexico, Canada (NAFTA)

$N = 4$: + China

Coalition coverage. $N = 5$: + India

$N = 6$: + Brazil

$N = 7$: + Korea

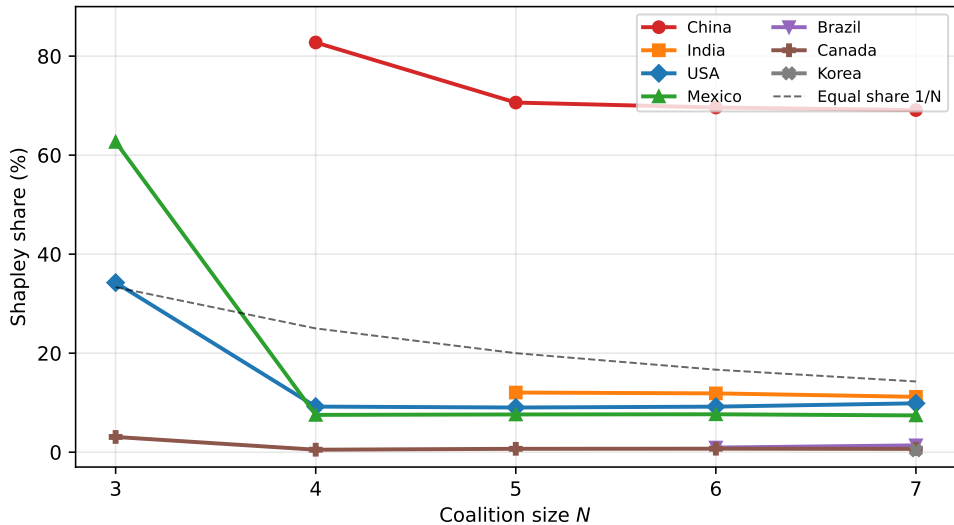
ROW handling. Excluded as Shapley player (would be a spurious size-jump from accounting, not strategic value); kept as passive GE participant. (Detail in paper Appendix A.)

Shapley shares are stable across N , dominated by China

	$N = 3$	$N = 4$	$N = 5$	$N = 6$	$N = 7$
USA	34.3%	9.2%	9.0%	9.2%	9.9%
Mexico	62.7%	7.5%	7.6%	7.7%	7.4%
Canada	3.1%	0.5%	0.7%	0.7%	0.7%
China	—	82.8%	70.6%	69.6%	69.1%
India	—	—	12.0%	11.9%	11.2%
Brazil	—	—	—	0.9%	1.4%
Korea	—	—	—	—	0.4%
$v(N)$ (\$B)	5.89	65.6	30.4	31.0	31.9

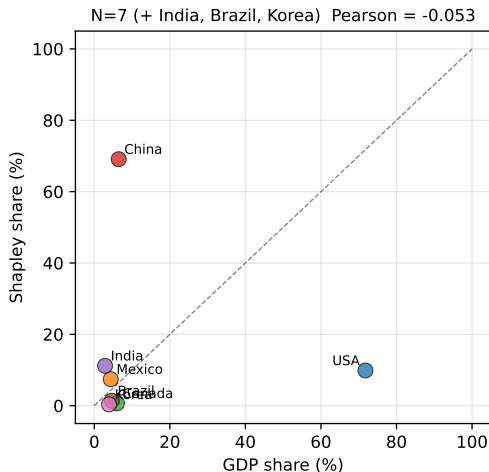
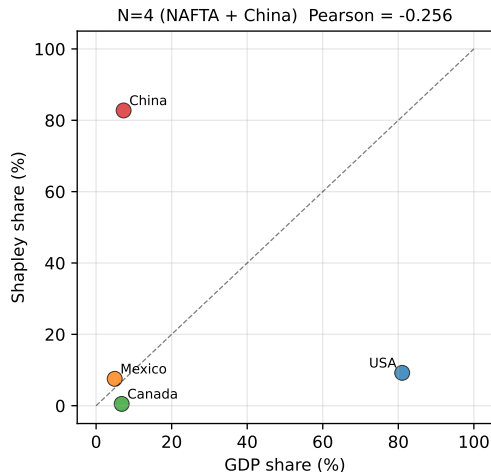
- China dominates $N \geq 4$ – not because it is large, but because applied tariff is high (32% avg).
- Canada, Brazil, Korea are *near-dummy* regardless of GDP.
- NAFTA's joint share collapses from 100% $\rightarrow$ 17% once China enters.

Same data, visually



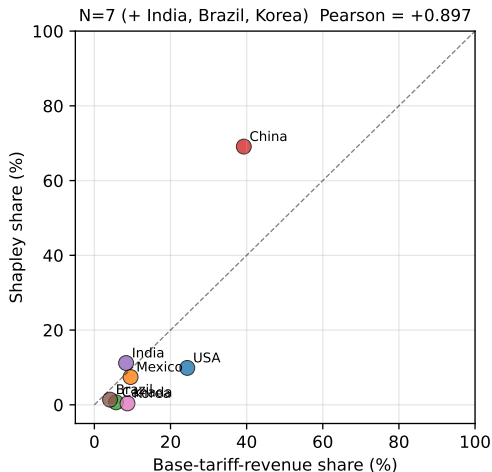
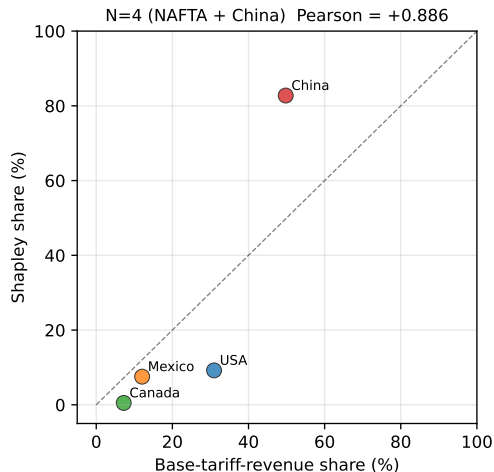
China's share is order of magnitude above any other player; existing NAFTA members' shares are

Shapley share is uncorrelated with GDP share



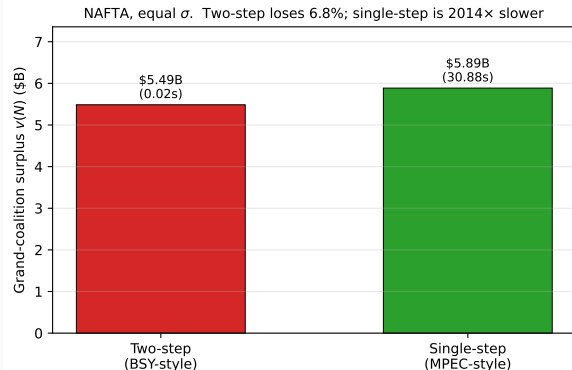
- $\text{Pearson} \in [-0.26, +0.01]$ across $N = 3, \dots, 7$.
- At $N = 4$: USA 81% GDP $\rightarrow$ 9% Shapley (premium -72 pp). China 7% GDP $\rightarrow$ 83% Shapley

Shapley share *is* correlated with tariff-revenue share



- Pearson rises sharply: $0.22 \rightarrow 0.88-0.90$ once $N \geq 4$.
- But **rankings still differ**: L^1 distance ≥ 0.56 ; Shapley puts India ahead of USA at $N \geq 5$, revenue

Two-step heuristic loses 6.8% of NAFTA surplus, silently



Why the gap?

- Two-step (BSY) drives $s = (0, 0, 0)$. *Full GE re-eval* $\Rightarrow$ Canada IR violated ($\widehat{W}_{\text{CAN}} = 1 - 8.5 \times 10^{-4}$).
- Single-step finds $s = (0.85, 0, 1.00)$: Canada must *not* fully liberalize because trade-share reallocation hurts it in GE.

Trade-off.

- Welfare loss $\sim 7\%$ at $N = 3$.
- Speed penalty $\sim 2014\times$.
- Expected to widen with N .

Robustness: three honest caveats

1. Game is not convex anywhere. 5/5 values of $N \in \{3, \dots, 7\}$ have systematic supermodularity violations (393 at $N = 7$, max gap \$4.5B). $\Rightarrow$ Shapley not guaranteed in core. (See backup.)

2. Threat-point endogeneity is first-order. NAFTA Shapley ranking flips: under 1993 applied, Mexico $>$ USA $>$ Canada. Under uniform $\tau = 50\%$, USA $>$ Canada $>$ Mexico. $v(N)$ ranges from \$0B to \$1,099B across three threat scenarios. (See backup.)

3. σ^* uniqueness: numerical only. Three Dirichlet-random starts at $N = 3$ converge to spread 2×10^{-2} in 5 iters; one of three meets strict tolerance. At $N = 4$: 2 of 3 trials done, consistent with uniqueness. $N \geq 5$ deferred (compute-prohibitive).

None of these are dispositive against the framework, but each is material; we report rather than hide.

What we do *not* claim (Manski 1995)

Identification. Distinct parameter values must produce distinct distributions of *observed* outcomes (Manski 1995). Of 2^N coalitional values, only $v(N)$ has a data analog (NAFTA 1993–2005). The remaining $2^N - 2$ are pure model output.

Four modeling choices that pin the imputation:

1. **Solution concept** – Shapley vs. Kalai-Smorodinsky vs. nucleolus vs. γ -core all yield different ϕ_i , none identified by data.
2. **Tariff lever** – scalar s_n , uniform across j, i . Real RTAs are sectoral.
3. **Threat point** – 1993 applied tariff, itself endogenous to 1947–93 GATT bargaining (Lima 2007).
4. **Non-member IR scope** – imposed on members only; maps to discriminatory PTAs without GATT XXIV reciprocity.

Reading. σ^* is a *computed structural imputation*, not a bargaining-power estimate. Honest scope $>$ overclaim.

Conclusion

Headline. Bargaining power = liberalization headroom, not market size.

- Shapley $\perp$ GDP (Pearson $\in [-0.26, +0.01]$).
- Shapley $\sim$ tariff revenue (Pearson ≥ 0.88 at $N \geq 4$), but rankings still differ.
- China dominates structurally with 69–83% Shapley share on 7% GDP share.

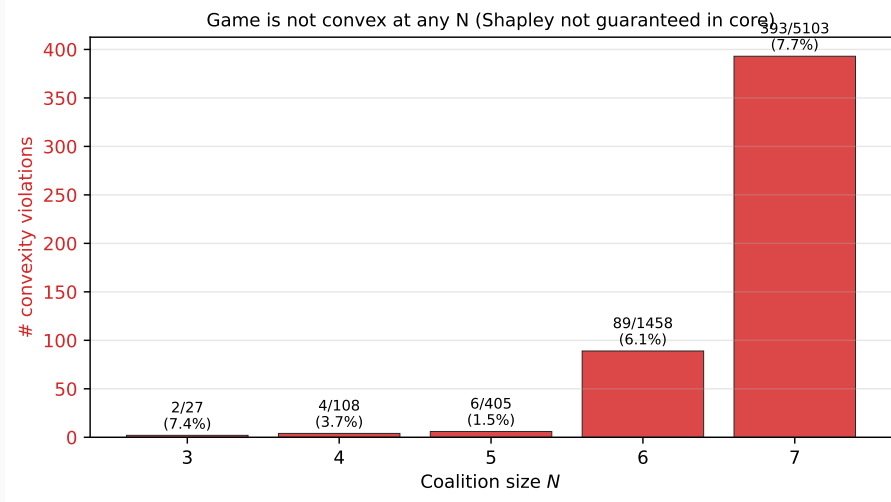
Methodological contributions.

- First multi-sector EK-CP $\times$ Shapley quantification at $N \leq 7$ exact.
- Single-step MPEC closes a 6.8% efficiency gap that BSY's two-step heuristic silently leaves on the table.

Open extensions.

- Sectoral lever (Pyomo+IPOPT MPEC) at $N = 31$.
- Alternative threat points (Smoot-Hawley 1947, prohibitive).
- Weighted Shapley with exogenous w_i (Kalai 1987) to nest BSY's asymmetric Nash.

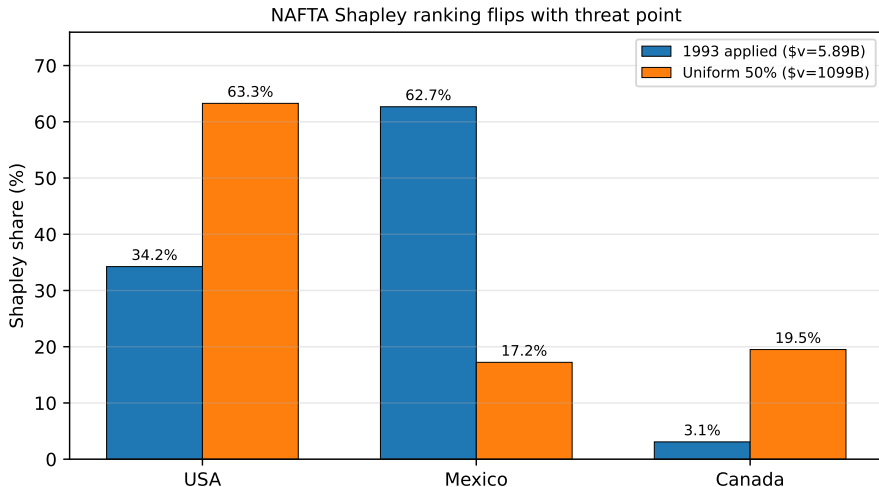
Backup: Convexity violations across N



Test. For every i and $S \subseteq T \subseteq N \setminus \{i\}$, check $v(T \cup \{i\}) - v(T) \geq v(S \cup \{i\}) - v(S)$.

Backup: Shapley violations: $2/27 = 7.4\%$, $4/108 = 3.7\%$, $6/405 = 1.5\%$, $89/1458 = 6.1\%$, $393/5103 = 7.7\%$

Backup: Threat point flips the NAFTA Shapley ranking



Threat point	$v(N)$ (\$B)	USA	Mexico	Canada
1993 applied	5.89	34.2%	62.7%	3.1%
Uniform 50%	1099	63.3%	17.2%	19.5%

Backup: Numerical evidence on σ^* uniqueness ($N = 3$)

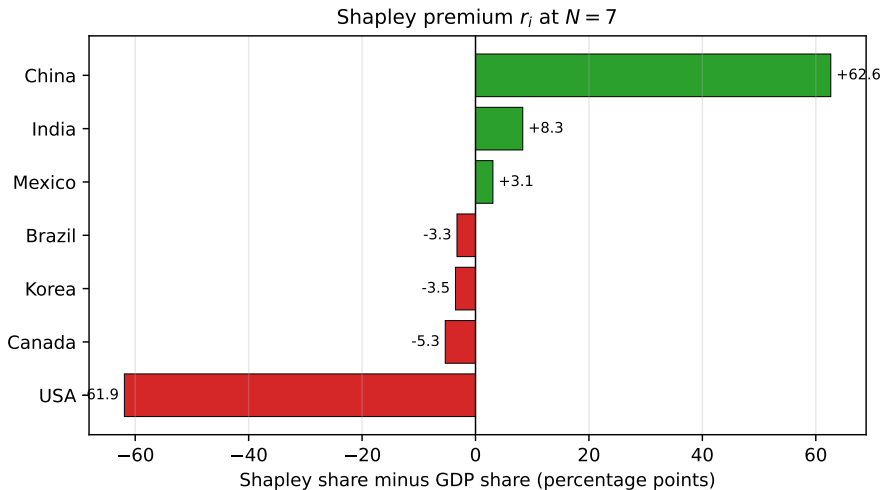
Test. 3 random Dirichlet starts on the 2-simplex; iterate $\Phi(\sigma) = \phi(v(\cdot \mid \sigma)) / \sum \phi$ to convergence.

	USA	Mexico	Canada	Iters
Trial 1	0.336	0.627	0.037	5
Trial 2	0.334	0.641	0.025	5 (converged)
Trial 3	0.340	0.622	0.038	
Consensus	0.337	0.630	0.033	—
Max spread	0.007	0.020	0.013	—

Verdict. 1 of 3 meets strict tolerance $\|\sigma^{(t)} - \sigma^{(t-1)}\|_\infty < 10^{-2}$ within 5 iters; the other two cluster within 2×10^{-2} of the converged trial. **Read as numerical evidence consistent with uniqueness, not formal proof.**

$N = 4$: 2 of 3 trials done (compute-bound, ~ 10 h each). $N \geq 5$ deferred.

Backup: Shapley premium $r_i = \phi_i / v(N) - \text{GDP}_i / \sum \text{GDP}$ at $N = 7$



Read. Premium = GE-implied Shapley share minus market-size benchmark. Sign tells the GDP relabel test verdict at the country level.

Backup: $N = 2$ nesting of BSY (Proposition 1)

Claim. With $N = \{i, j\}$ and singleton normalization $v(\{i\}) = v(\{j\}) = 0$, the Shapley fixed-point weights satisfy $\sigma_i^* = \sigma_j^* = \frac{1}{2}$, and the program reduces to the symmetric Nash bargain.

Proof sketch. For $N = 2$, the Shapley formula gives

$$\phi_i = \frac{1}{2} v(\{i\}) + \frac{1}{2} [v(\{i, j\}) - v(\{j\})].$$

With $v(\{i\}) = v(\{j\}) = 0$, $\phi_i = \phi_j = \frac{1}{2} v(\{i, j\})$, so $\sigma_i^* = \phi_i / (\phi_i + \phi_j) = \frac{1}{2}$. $\square$

Caveat. The asymmetric BSY case ($\mu_i \neq \frac{1}{2}$, GMM-estimated from observed RTAs) is *not* nested by Shapley alone; it requires the weighted Shapley value of Kalai (1987) with exogenous w_i .

Implication. At $N = 2$ our framework is observationally equivalent to the symmetric Nash bargain regardless of country characteristics. The marginal value-add of Shapley appears only at $N \geq 3$ via cross-coalition heterogeneity in $v(S)$.

Backup: Full MPEC formulation

$$\begin{aligned}
 & \max_{\substack{\{s_n\}_{n \in S}, \\ \widehat{w}, \widehat{P}, \widehat{C}, \pi, X'}}} \sum_{i \in S} \sigma_i \log(\widehat{W}_i - 1) \\
 & \text{s.t. } \widehat{c}_n^j = \widehat{w}_n^{j,n} \prod_k (\widehat{P}_n^k)^{\gamma_n^{k,j}} \quad \forall n, j \\
 & \quad \widehat{P}_n^j = \left[\sum_i \pi_{ni}^j \left(\widehat{\kappa}_{ni}^j(s_n) \widehat{c}_i^j \right)^{-\theta^j} \right]^{-1/\theta^j} \quad \forall n, j \\
 & \quad \widehat{\pi}_{ni}^j = \left(\widehat{c}_i^j \widehat{\kappa}_{ni}^j(s_n) / \widehat{P}_n^j \right)^{-\theta^j} \quad \forall n, i, j \\
 & \quad X_n^{j'} = \sum_k \gamma_n^{j,k} \sum_i \frac{\pi_{in}^{k'} X_i^{k'}}{1 + \tau_{in}^{k'}} + \alpha_n^j I_n^{j'} \quad \forall n, j \\
 & \quad \sum_{j,i} \frac{\pi_{ni}^{j'} X_n^{j'}}{1 + \tau_{ni}^{j'}} - D_n = \sum_{j,i} \frac{\pi_{in}^{j'} X_i^{j'}}{1 + \tau_{in}^{j'}} \quad \forall n \\
 & \quad \widehat{\kappa}_{ni}^j(s_n) = (1 + s_n \tau_{ni}^{j,0}) / (1 + \tau_{ni}^{j,0}) \quad \forall n, i, j \\
 & \quad \widehat{W}_i \geq 1 + \epsilon, \quad \forall i \in S \quad (\epsilon = 10^{-6}) \\
 & \quad s_n \in [0, 1] \quad \forall n \in S; \quad s_n = 1 \quad \forall n \notin S.
 \end{aligned}$$

Decision count: $|S|$ outer scalars + $\sim N(1 + 3J + NJ)$ equilibrium variables. One trade-balance constraint redundant by Walras' law (dropped).

Current implementation: SLSQP outer, $\times$ hand coded hot algebra inner (2 step post). Pyomo + IPOPT

Backup: Two notions of IR, both verified ex post

Bargaining IR (per coalition).

Inside each $v(S)$ MPEC, we enforce $\widehat{W}_i \geq 1 + \epsilon$ for all $i \in S$. Verified active in solver output.

Imputation IR (Shapley non-negativity).

We require $\phi_n \geq v(\{n\}) = 0$ for all n at the converged σ^* . Verified ex post via the realized Shapley table: no negative shares observed at $N = 3, \dots, 7$.

Distinct concepts.

- Bargaining IR is enforced *inside* each MPEC.
- Imputation IR is a property of ϕ and follows from monotonicity of v in coalition inclusion (Condition (iii) in our existence conjecture).

Both hold in our application even though core membership is not guaranteed by theory (game is not convex).